W08 Pre-lab Quiz: DAC and DMA

Started: Feb 21 at 11:41am

Quiz Instructions

Week 08

Pre-Lab Quiz

Review all study material and answer the questions.

Question 1 0.25 pts

In a typical DMA transfer to a peripheral, what are the source and destination addresses?

- ○ Peripheral source and memory destination
- ○ Constant source and register destination
- ◉ Memory source and peripheral destination
- ○ Stack source and peripheral destination
- ○ memory
- ○ Register source and memory destination

Question 2 0.25 pts

DMA-controlled transfers have to be assigned a priority because:

- ○ The CPU cannot access memory during DMA transfers
- ○ Priorities determine the transfer speed of each channel
- ○ DMA transfers always require the highest system priority

- [ ] Only one DMA channel can be active at a time
- [x] Multiple DMA channels may request access to the bus at the same time

Question 3 0.25 pts

Is it necessary to use a timer to trigger the ADC (Analog-to-Digital Converter) sampling rate in a microcontroller?

- [x] No
- [ ] Yes, always
- [ ] Only when using DMA
- [ ] Only if the ADC is in continuous mode
- [ ] Only for high-speed sampling

Question 4 0.25 pts

How many digital-to-analog converter (DAC) channels are available on the STM32-Nucleo-L476RG development board?

- [ ] 4
- [ ] 1
- [ ] None
- [x] 2
- [ ] 8

Question 5 0.25 pts

A software-triggered ADC sample is usually the best method to maximize sampling throughput.

- [ ] True
- [x] False

Question 6 0.25 pts

Pure, singular-frequency sine-waves are found everywhere in nature.

- ○ True
- ◉ False

Question 7 0.25 pts

Which of the following is **not** a valid DMA transfer mode supported by the built-in DMA controllers in STM32 microcontrollers?

- ○ Memory-to-Peripheral transfers
- ◉ Burst Mode
- ○ Circular Mode
- ○ Memory-to-Memory transfers
- ○ Peripheral-to-Memory transfers
- ○ Normal Mode

Question 8 0.25 pts

A square wave pulse in the time domain can be decomposed into a sinc function (y = sin(x)/x) in the frequency domain.

- ◉ True
- ○ False

Question 9 0.25 pts

In converting an analog signal to digital via sampling, the minimum sampling frequency needs to be:

- ○ At least twice of any frequency component of the signal
- ○ At least twice the lowest frequency component of the signal

- ○ At least half the highest frequency component of the signal
- ○ At least twice the average frequency of the signal
- ◉ At least twice the highest frequency component of the signal

Question 10 0.25 pts

What is NOT a reason to use digital computing systems.

- ○ They can store and manipulate large amounts of data
- ○ They can be easily reprogrammed for different tasks
- ◉ They naturally process continuous analog signals without conversion
- ○ They are scalable and can handle complex operations
- ○ They are reliable and repeatable

Question 11 0.25 pts

A waveform generated from digitized samples that incorrectly represents the original because of an inadequate sample rate is known as:

- ○ Clipped
- ◉ Aliased
- ○ Filtered
- ○ Quantized
- ○ Saturated

Question 12 0.25 pts

DMA controllers can be configured for circular transfers and can issue interrupts each time the buffer is filled.

- ◉ True

False

Question 13 1 pts

What is the effect of applying a moving average filter multiple times (multiple-pass)?

It sharpens the step response

It reduces the filter's execution speed significantly

It introduces nonlinear phase distortion

It eliminates the need for convolution

It improves stopband attenuation and shapes the filter kernel toward a Gaussian

Question 14 1 pts

What is the primary trade-off when using a longer moving average filter?

Improved noise reduction at the cost of reduced edge sharpness

Sharper frequency cutoff with increased phase distortion

Increased computational complexity with no benefit

Improved stopband attenuation with nonlinear phase

Reduced DC gain and increased ripple

Question 15 1 pts

What is the signal-to-noise ratio (SNR) in statistical terms?

The inverse of the coefficient of variation

The standard deviation divided by the mean

The mean divided by the standard deviation

○
The square of the mean

○
The ratio of peak-to-peak amplitude to variance

Question 16 1 pts

What does the Central Limit Theorem imply for digital noise generation?

○
The histogram of any signal is Gaussian

○
The average of random variables is always zero

◉
The sum of many random variables tends toward a Gaussian distribution

○
The mean of a signal is equal to its median

○
The variance of a signal is always constant

Question 17 1 pts

What is the main disadvantage of using Gaussian and Blackman filters compared to the moving average filter?

○
They cannot be implemented recursively

○
They are unstable for long signals

○
They introduce nonlinear phase distortion

◉
They require convolution and are computationally slower

○
They have poor stopband attenuation

Question 18 1 pts

What is the main advantage of using a recursive implementation of the moving average filter?

○
It provides better frequency selectivity

○
It introduces beneficial phase shifts

- ○ It eliminates the need for a filter kernel
- ○ It allows for non-linear filtering
- ◉ It requires only one addition and one subtraction per output sample

Question 19 1 pts

What is the purpose of the transition band in a filter's frequency response?

- ○ It indicates the filter's gain at DC
- ◉ It defines the range between the passband and stopband
- ○ It shows the filter's phase delay
- ○ It determines the filter's impulse response length
- ○ It measures the filter's risetime

Quiz saved at 12:11pm Submit Quiz

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